

Koa Shen

Mechanical Engineering Intern Candidate · Vehicle Hardware · Manufacturing · Test

(415) 810-8344 koashensf@gmail.com linkedin.com/in/koa-shen/ github.com/koa-shen koashen.com

EDUCATION

University of California, Santa Barbara

Expected June 2028

B.S. Mechanical Engineering · 3.97 major / 3.94 overall GPA · Dean's Honors every quarter

Santa Barbara, CA

- Coursework: Dynamics, Mechanics of Materials, Circuits. Current: Dynamical Systems and Machine Learning.

EXPERIENCE

Fabrication Team Lead

Fall 2024 – Present

Gaucha Racing (FSAE EV) — UC Santa Barbara

Santa Barbara, CA

- Led 15 teammates through driver-safety hardware design, fabrication, and installation; GR26 passed FSAE technical inspection fully rules-compliant.
- CNC-machined 33 suspension components from 4130 chromoly steel and 7075-T6 aluminum on Haas mills and lathes; held bearing fits within 0.0005 in.
- Own chassis and ergonomics packaging, coordinating welded-tube structures, assembly interfaces, and manufacturing with vehicle subteams.
- Placed 2nd among UC teams at the 2026 Michigan FSAE Electric competition and 4th among California teams.

Mechanical Engineering Intern

Summer 2026 – Present (extended to part-time remote)

SLAC National Accelerator Laboratory — LCLS

Menlo Park, CA

- Built an open-source Python/Drake digital twin for a 30-DOF, 305-body beamline assembly, with CAD import, collision geometry, and millisecond checks that exposed at least five pinch points.
- Created 15 reusable motion-stage models for three LCLS assemblies; applied simulation findings and instrument-scientist feedback to redesign optics packaging.
- Produced Solid Edge designs and PDM-controlled documentation with GD&T; checked mount loads to address crash and drop risks around \$100k+ detectors.

Maintenance Technician

Summer 2025

Exploratorium

San Francisco, CA

- Serviced 100+ public exhibits and led three major overhauls, reverse-engineering mechanisms and fabricating replacement hardware.
- Redesigned a load-spreading bezel to stop acrylic enclosure cracking, avoiding \$3,000+ in replacement costs; documented CAD, CAM, and remanufacturing notes in Confluence.

SELECTED PROJECTS

Compact Cycloidal Reducer & Stepper Package

Summer 2026 – Present

- Designed and fabricated a 15:1 cycloidal reducer sized within a NEMA 17 motor footprint; selected printed-material orientations for load capacity and dimensional accuracy.
- Built C++ firmware for TMC2209 stepper control and AS5600 encoder feedback; bench-validated position correction and missed-step detection.

FSAE Steering Wheel Development

Fall 2024 – Spring 2026

- Integrated buttons, potentiometers, display, and quick release into an FSAE steering wheel; used FEA-guided iterations to reduce assembly mass from 5 lb to 1.7 lb while maintaining driver access and egress.

TECHNICAL SKILLS

Mechanical Design & Analysis: SolidWorks, Solid Edge, GD&T, FEA, Statics/dynamics, Tolerance-aware design

Manufacturing & Prototyping: CNC/manual machining, Mastercam, Welding, 3D printing, 4130 steel, 7075-T6 aluminum

Controls & Simulation: C/C++, Python, MATLAB, Drake kinematics, Collision detection, Stepper/encoder systems